Primary reason recorded for efficacy failure in the 27 drug classes whose target engagement was confirmed in patients

Target hypothesis: the target and the direction of the intended effect — n = 16

- ◆ Not causal: output moved, disease did not (12): H3 antagonists · AD; 5-HT6 antagonists · AD; anti-IL-13 · asthma; anti-IL-33 · atopic eczema; p38 inhibitors · COPD; HCAR2 agonists · lipids; HDAC inhibitors · MDS; anti-IGF-1R · NSCLC; PDE10A inhibitors · SCZ; GPR119 agonists · T2D; 11β-HSD1 inh. · T2D; MEK inhibitors · pancreas
- ◆ Causal at another disease stage (2): BACE1 inhibitors · AD; CDK4/6 inhibitors · breast
- ◆ Causal only in a subgroup (0): *none*
- Causal but compensated or bypassed (1): anti-EGFR (new) · CRC
- Causal, effect too small (1): pan-PI3K inhibitors · breast

Modulation: the drug, its dose, or the site of engagement — n = 6

- Insufficient degree or duration of engagement (6): HSP90 inhibitors · breast; PI3K/mTOR inh. · breast; NRTIs (new) · HIV; anti-GM-CSF · RA; BTK inhibitors · RA; proteasome inh. · myeloma

Trial: endpoint, population, or treatment context — n = 4

- Population or placebo response (3): allergen peptides · allergy; allergen IT · rhinitis; anti-CGRP · migraine
- Combination or comparator context (1): DP2 antagonists · asthma

Not stated in the record — n = 1

- Not stated in the record (1): DP2 antagonists · rhinitis

class never approved in this disease

class approved in this disease; the contradicted entry concerns a stage, a population, or one member

◆ category present in the current manuscript draft; the remaining categories are added here